INCH-POUND

MS24264T
1 December 2016
SUPERSEDING
MS24264R
20 August 2009

DETAIL SPECIFICATION SHEET

# CONNECTORS, RECEPTACLE, ELECTRICAL, FLANGE MOUNTING, MINIATURE, CLASSES E, F, G AND R

Inactive for new design for 14 November 1977. For new design, use MIL-DTL-83723, series III, which is interchangeable.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-26500.

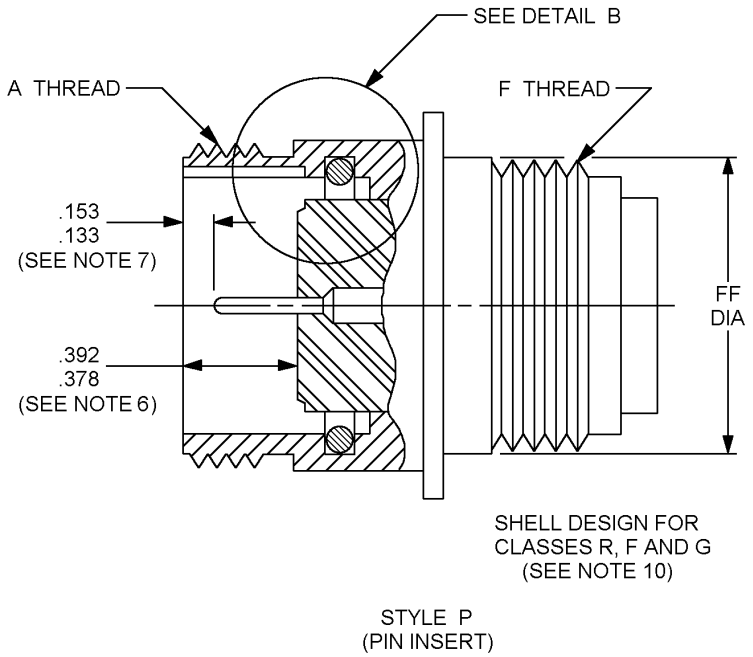
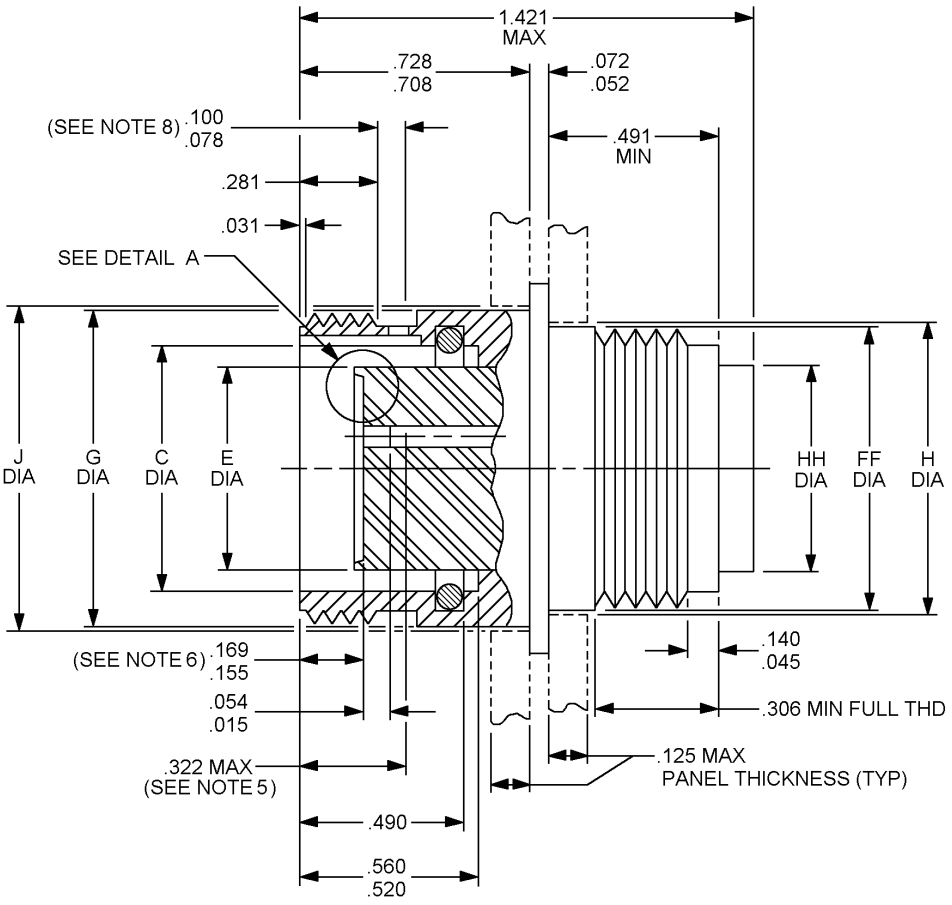



FIGURE 1. Receptacle, threaded, for classes F, G and R.

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.




STYLE S
(SOCKET INSERT)

FIGURE 1. Receptacle, threaded, for classes F, G and R – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

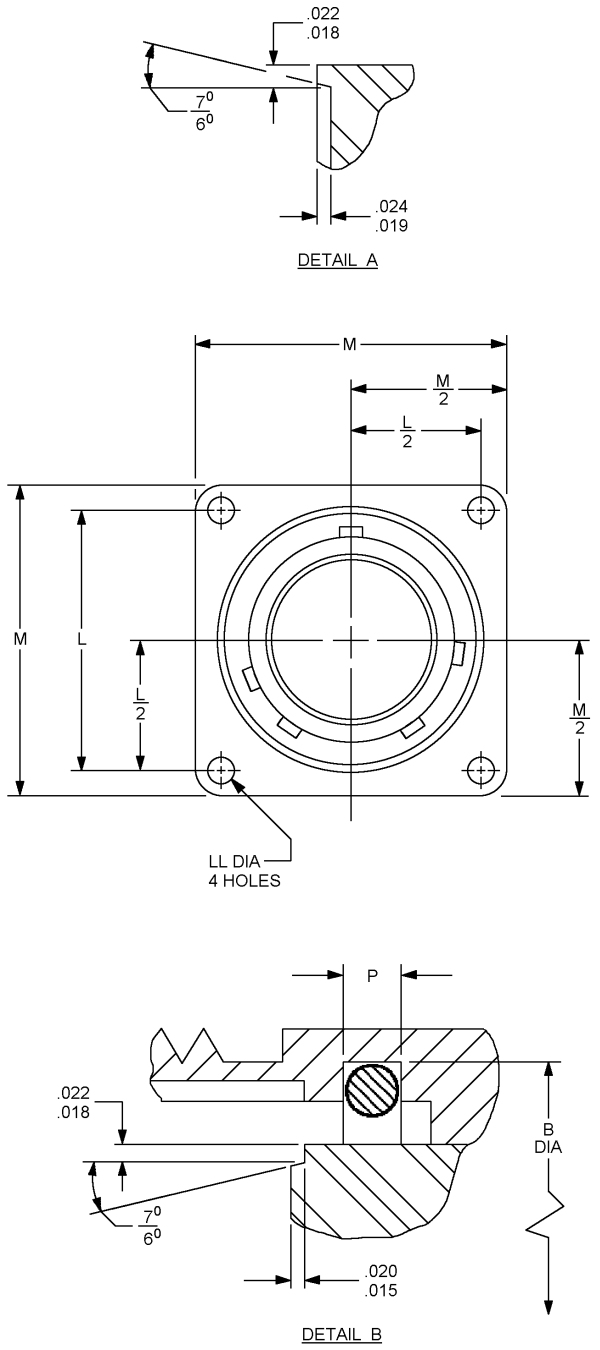



FIGURE 1. Receptacle, threaded, for classes F, G and R – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .015 | 0.38 | .052 | 1.32 | .153 | 3.89 | .490 | 12.45 |
| .018 | 0.46 | .054 | 1.37 | .155 | 3.94 | .491 | 12.47 |
| .019 | 0.48 | .072 | 1.83 | .169 | 4.29 | .520 | 13.21 |
| .020 | 0.51 | .078 | 1.98 | .281 | 7.14 | .560 | 14.22 |
| .022 | 0.56 | .100 | 2.54 | .306 | 7.77 | .708 | 17.98 |
| .024 | 0.61 | .125 | 3.18 | .322 | 8.18 | .728 | 18.49 |
| .031 | 0.79 | .133 | 3.38 | .378 | 9.60 | 1.421 | 36.09 |
| .045 | 1.14 | .140 | 3.56 | .392 | 9.96 | | |

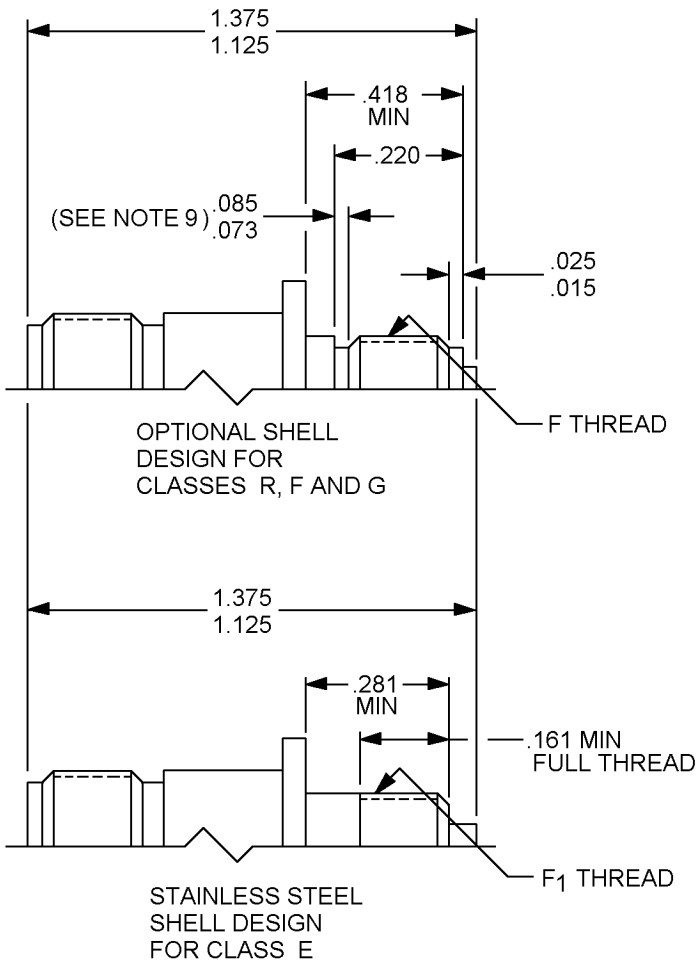
NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances on decimals is ± .005.
2. Metric equivalents are given for information only.
3. Use tool MIL-I-81969/17 to assemble contacts into this connector, and use tool MIL-I-81969/19 to remove contacts from this connector.
4. All diameters to be concentric with each other within .015 TIR. All diameters in the same plane to be concentric with each other within .004 TIR.
5. Distance between end of shell and the point at which a gauge pin having the same basic diameter as the mating contact and a square face, engages socket contact spring.
6. Dimensions on pin and socket contact locations and end of shell to insert faces apply when contacts are placed in inserts for inspection or application.
7. Dimension .133 may reduce to .118 minimum under pressures caused by molded cable assemblies or sharp cable bends.
8. Thread relief groove is optional on shell. When groove is omitted the length of full thread from front of shell will be .310 minimum and thread run out .385 maximum.
9. Thread relief groove is optional on shell. When groove is omitted, the length of full thread from rear of shell will be .221 minimum.
10. Environment resistant (classes F and R) receptacles, type T aluminum shell material. Grounding environment resistant (class G) receptacles, type T aluminum shell material. Environment resistant (class E) receptacles, type T stainless steel shell material. These receptacles mate with plug MS24266 type T.
11. True position (TP) tolerances specified are for maximum material conditions (MMC).

FIGURE 1. Receptacle, threaded, for classes F, G and R – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .015 | 0.38 | .220 | 5.59 |
| .025 | 0.63 | .281 | 7.14 |
| .073 | 1.85 | .418 | 10.62 |
| .086 | 2.18 | 1.125 | 28.57 |
| .161 | 4.09 | 1.375 | 34.92 |

FIGURE 2. Receptacle, optional shell design for classes F, G, R and stainless steel shell design for class E.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances on decimals is ± .005.
2. Metric equivalents are given for information only.
3. Use tool MIL-I-81969/17 to assemble contacts into this connector, and use tool MIL-I-81969/19 to remove contacts from this connector.
4. All diameters to be concentric with each other within .015 TIR. All diameters in the same plane to be concentric with each other within .004 TIR.
5. Distance between end of shell and the point at which a gauge pin having the same basic diameter as the mating contact and a square face, engages socket contact spring.
6. Dimensions on pin and socket contact locations and end of shell to insert faces apply when contacts are placed in inserts for inspection or application.
7. Dimension .133 may reduce to .118 minimum under pressures caused by molded cable assemblies or sharp cable bends.
8. Thread relief groove is optional on shell. When groove is omitted the length of full thread from front of shell will be .310 minimum and thread run out .385 maximum.
9. Thread relief groove is optional on shell. When groove is omitted, the length of full thread from rear of shell will be .221 minimum.
10. Environment resistant (classes F and R) receptacles, type T - aluminum shell material. Grounding environment resistant (class G) receptacles, type T - aluminum shell material. Environment resistant (class E) receptacles, type T - stainless steel shell material. These receptacles mate with plug MS24266, type T.
11. True position (TP) tolerances specified are for maximum material conditions (MMC).

FIGURE 2. Receptacle, optional shell design for classes F, G, R and stainless steel shell design for class E – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

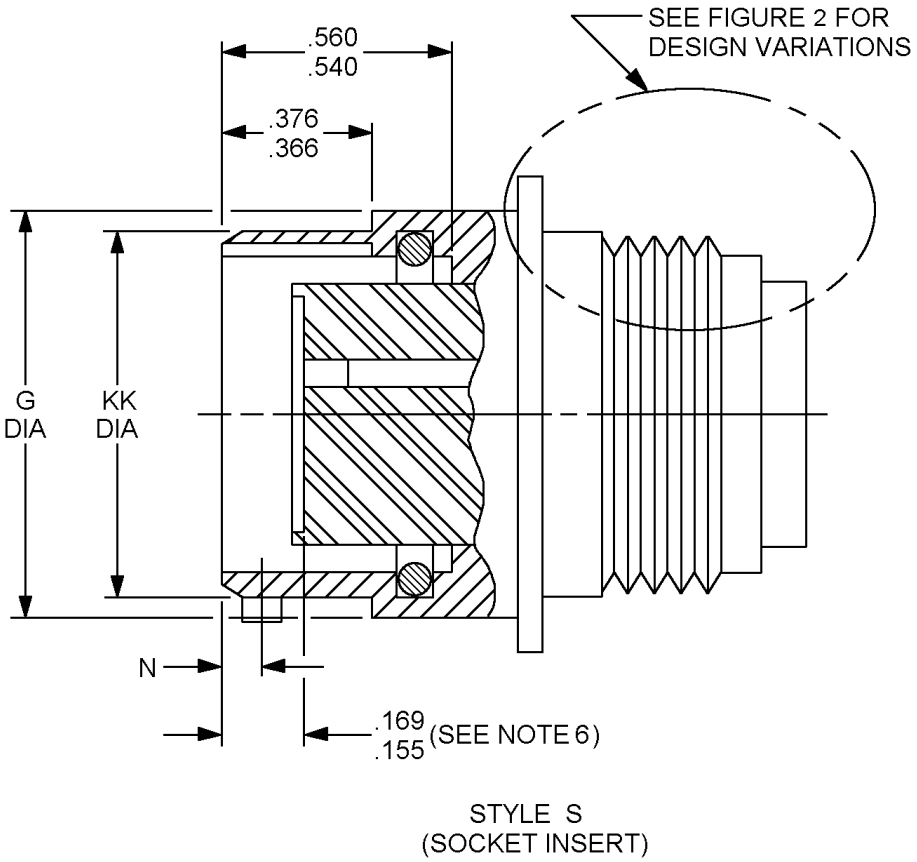



FIGURE 3. Receptacle, bayonet, optional design.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

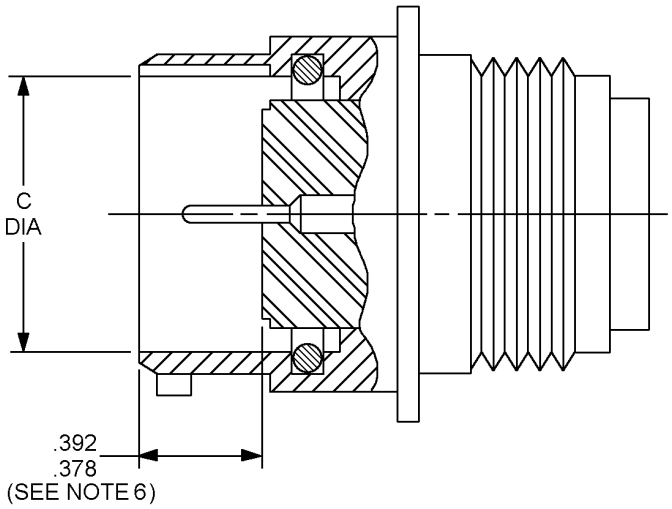
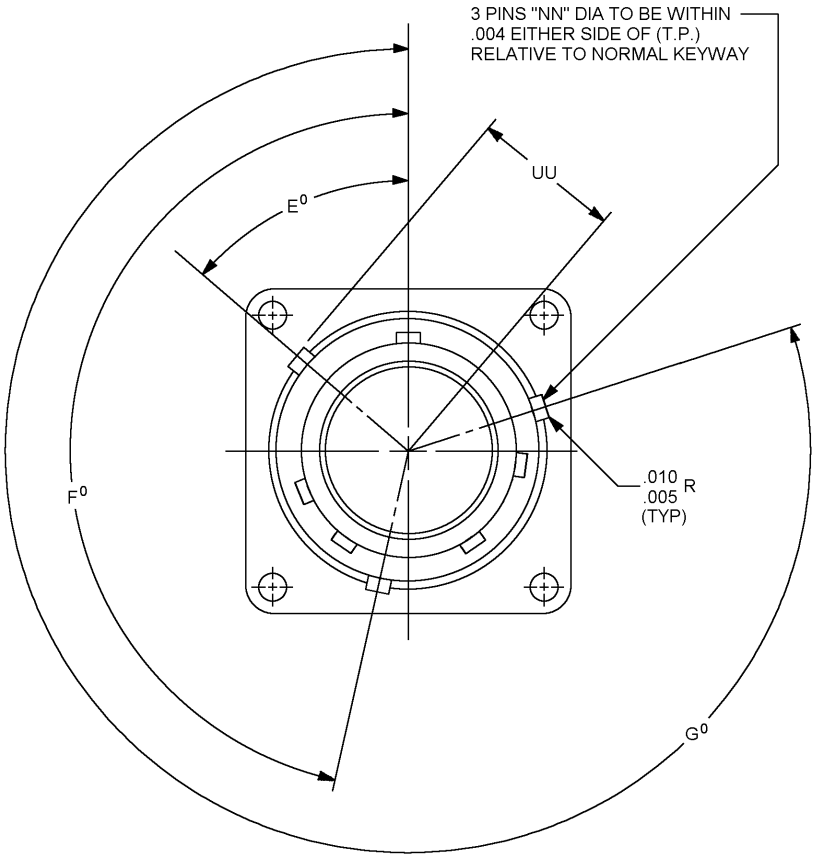



FIGURE 3. Receptacle, bayonet, optional design - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .004 | 0.10 | .376 | 9.55 |
| .005 | 0.13 | .378 | 9.6 |
| .010 | 0.25 | .392 | 9.96 |
| .155 | 3.94 | .540 | 13.72 |
| .169 | 4.29 | .560 | 14.22 |
| .368 | 9.35 | | |

FIGURE 3. Receptacle, bayonet optional design - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances on decimals is ± .005.
2. Metric equivalents are given for general information only.
3. Use tool MIL-I-81969/17 to assemble contacts into this connector, and use tool MIL-I-81969/19 to remove contacts from this connector.
4. All diameters to be concentric with each other within .015 TIR. All diameters in the Same plane to be concentric with each other within .004 TIR.
5. Distance between end of shell and the point at which a gauge pin having the same basic diameter as the mating contact and a square face, engages socket contact spring.
6. Dimension on pin and socket contact locations and end of shell to insert faces apply when contacts are placed in inserts for inspection or application.
7. Dimension .133 may reduce to .118 minimum under pressures caused by molded cable assemblies or sharp cable bends.
8. Environment resistant (classes F and R) receptacles, type B - aluminum shell material. Grounding environment resistant (class G) receptacles, type B - aluminum shell material. Environment resistant (class E) receptacles, type B - stainless steel shell material. This receptacle mates with plug MS24266, type B.
9. True position (TP) tolerances specified are for maximum material conditions (MMC).

FIGURE 3. Receptacle, bayonet optional design - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Shell size | A UNEF-2A coupling | B dia | C dia + .005 (0.13) - .000 (0.00) | E max Insert dia | F UNEF-2A access |
|---|---|---|---|---|---|
| 8 | .5625-24 (14.27) | .508 (12.90) | .428 (10.87) | .318 (8.08) | .4375-28 (11.10) |
| 10 | .6875-24 (17.45) | .636 (16.15) | .530 (13.46) | .394 (10.01) | .5625-24 (14.27) |
| 12 | .875-20 (22.22) | .806 (20.47) | .700 (17.78) | .564 (14.33) | .750-20 (19.05) |
| 14 | .9375-20 (23.80) | .875 (22.22) | .769 (19.53) | .633 (16.08) | .8125-20 (20.62) |
| 16 | 1.0625-18 (26.97) | 1.002 (25.45) | .896 (22.76) | .760 (19.30) | .9375-20 (23.80) |
| 18 | 1.1875-18 (30.15) | 1.108 (28.62) | 1.002 (25.45) | .866 (22.00) | 1.0625-18 (26.97) |
| 20 | 1.3125-18 (33.32) | 1.233 (31.32) | 1.127 (28.62) | .991 (25.17) | 1.1875-18 (30.15) |
| 22 | 1.4375-18 (36.50) | 1.358 (34.49) | 1.252 (31.80) | 1.116 (28.35) | 1.3125-18 (33.32) |
| 24 | 1.5625-18 (39.67) | 1.483 (37.67) | 1.377 (34.98) | 1.241 (31.52) | 1.4375-18 (36.50) |

FIGURE 4. Receptacle dimensions.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Shell size | $F_1$ -36NS-2A | $F_1$ pitch dia | FF max dia | FF min dia (see note 3) | G max dia |
|---|---|---|---|---|---|
| 8 | .4340 (11.024) | .4151/.4114 (10.544/10.450) | .437 (11.10) | .428 (10.87) | .561 (14.25) |
| 10 | .5634 (14.310) | .5454/.5415 (13.853/13.754) | .562 (14.27) | .554 (14.07) | .696 (17.68) |
| 12 | .7334 (18.628) | .7154/.7115 (18.171/18.072) | .750 (19.50) | .728 (18.49) | .875 (22.22) |
| 14 | .8032 (20.401) | .7841/.7806 (19.961/19.827) | .812 (20.62) | .796 (20.22) | .935 (23.75) |
| 16 | .9302 (23.627) | .9110/.9074 (23.134/23.048) | .938 (23.82) | .923 (23.44) | 1.062 (26.97) |
| 18 | 1.0362 (26.319) | 1.0171/1.0134 (25.834/25.740) | 1.062 (26.97) | 1.029 (26.14) | 1.187 (30.15) |
| 20 | 1.1611 (29.492) | 1.1431/1.1385 (29.034/28.918) | 1.182 (30.02) | 1.156 (29.36) | 1.312 (33.32) |
| 22 | 1.2862 (32.669) | 1.2670/1.2633 (32.182/32.088) | 1.312 (33.32) | 1.279 (32.49) | 1.437 (36.50) |
| 24 | 1.4111 (35.842) | 1.3931/1.3885 (35.385/35.268) | 1.432 (36.37) | 1.406 (35.71) | 1.562 (39.67) |

FIGURE 4. Receptacle dimensions – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Shell size | H min dia front mounting hole | HH max grommet dia | J min dia back mounting hole | KK dia + .000 (0.00) - .005 (0.13) | L mounting hole | LL dia + .000 (0.00) - .009 (0.23) |
|---|---|---|---|---|---|---|
| 8 | .447 (11.35) | .328 (8.33) | .620 (15.75) | .536 (13.61) | .594 (15.09) | .125 (3.18) |
| 10 | .572 (14.53) | .420 (10.67) | .748 (19.00) | .659 (16.74) | .719 (18.26) | .125 (3.18) |
| 12 | .760 (19.30) | .580 (14.73) | .913 (23.19) | .829 (21.06) | .812 (20.62) | .125 (3.18) |
| 14 | .822 (20.88) | .664 (16.86) | .980 (24.90) | .898 (22.81) | .906 (23.01) | .125 (3.18) |
| 16 | .948 (24.08) | .769 (19.53) | 1.107 (28.12) | 1.025 (26.03) | .969 (24.61) | .125 (3.18) |
| 18 | 1.072 (28.12) | .902 (22.91) | 1.209 (30.71) | 1.131 (28.73) | 1.062 (26.97) | .125 (3.18) |
| 20 | 1.192 (30.28) | 1.033 (26.24) | 1.337 (33.96) | 1.256 (31.90) | 1.156 (29.36) | .125 (3.18) |
| 22 | 1.322 (33.58) | 1.152 (29.26) | 1.452 (36.89) | 1.381 (35.08) | 1.250 (31.75) | .125 (.318) |
| 24 | 1.442 (36.63) | 1.282 (32.56) | 1.577 (40.055) | 1.505 (38.25) | 1.375 (34.925) | .154 (3.91) |

FIGURE 4. Receptacle dimensions – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Shell size | M length side | N + .000 (0.00) - .004 (0.10) | NN + .000 (0.00) - .004 (0.10) | P + .000 (0.00) - .005 (0.13) | UU + .000 (0.00) + .009 (0.23) |
|---|---|---|---|---|---|
| 8 | .812 (20.62) | .101 (2.56) | .078 (1.93) | .076 (1.93) | .310 (7.87) |
| 10 | .937 (23.80) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .374 (9.50) |
| 12 | 1.031 (26.19) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .459 (11.66) |
| 14 | 1.125 (28.58) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .494 (12.55) |
| 16 | 1.250 (31.76) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .557 (14.15) |
| 18 | 1.343 (34.11) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .610 (15.49) |
| 20 | 1.437 (36.50) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .673 (17.09) |
| 22 | 1.562 (39.67) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .735 (18.67) |
| 24 | 1.703 (43.26) | .094 (2.39) | .093 (2.36) | .089 (2.26) | .798 (20.27) |

| Position | Shell size 8 | | | Shell size 10 | | | Shell sizes 12 thru 24 | | |
|---|---|---|---|---|---|---|---|---|---|
| | E ° | F° | G° | E° | F° | G° | E° | F° | G° |
| Normal | 60 | 180 | 300 | 60 | 180 | 300 | 50 | 170 | 290 |
| 6 | 46 | 166 | 286 | 46 | 166 | 286 | 50 | 170 | 290 |
| 7 | 46 | 166 | 286 | 46 | 166 | 286 | 50 | 170 | 290 |
| 8 | 60 | 180 | 300 | 60 | 180 | 300 | 50 | 170 | 290 |
| 9 | 89 | 209 | 329 | 89 | 209 | 329 | 50 | 170 | 290 |
| Y | - | - | - | 60 | 180 | 300 | 50 | 170 | 290 |

FIGURE 4. Receptacle dimensions – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Metric equivalent are given for information only.
3. The accessory threads may be formed with an optional modified major diameter. The following details apply: The major diameter of the accessory thread may be modified by truncation. The truncated major diameter shall be greater than FF min. The width of the flat across the top of the thread may increase as the truncation approaches FF min. All other features of the accessory thread, including the theoretical crest apex and the flank angle, shall meet standard thread form limits. The truncation of the major diameter shall not be less than the pitch diameter.

FIGURE 4. Receptacle dimensions – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Weight chart | | |
|---|---|---|
| Maximum connector weight in lbs | | |
| Pin insert | | |
| MS PIN | Less contacts | With contacts |
| MS24264R8T2PN | .023 (0.58) | .025 (0.64) |
| MS24264R10T2PN | .039 (0.99) | .041 (1.04) |
| MS24264R10T5PN | .038 (0.96) | .042 (1.07) |
| MS24264R10T20PN | .038 (0.96) | .042 (1.07) |
| MS24264R12T3PN | .053 (1.35) | .058 (1.47) |
| MS24264R12T12PN | .051 (1.30) | .061 (1.55) |
| MS24264R14T3PN | .057 (1.48) | .067 (1.70) |
| MS24264R14T4PN | .057 (1.48) | .072 (1.83) |
| MS24264R14T7PN | .057 (1.48) | .070 (1.78) |
| MS24264R14T12PN | .057 (1.48) | .070 (1.78) |
| MS24264R14T15PN | .057 (1.48) | .071 (1.80) |
| MS24264R16T10PN | .067 (1.70) | .085 (2.16) |
| MS24264R16T24PN | .067 (1.71) | .087 (2.21) |
| MS24264R18T8PN | .081 (2.06) | .110 (2.79) |
| MS24264R18T11PN | .079 (2.01) | .103 (2.62) |
| MS24264R18T14PN | .080 (2.03) | .105 (2.67) |
| MS24264R18T31PN | .079 (2.01) | .105 (2.67) |
| MS24264R20T16PN | .095 (2.41) | .123 (3.12) |
| MS24264R20T25PN | .092 (2.34) | .130 (3.30) |
| MS24264R20T28PN | .093 (2.36) | .128 (3.25) |
| MS24264R20T39PN | .093 (2.36) | .128 (3.25) |
| MS24264R20T41PN | .093 (2.36) | .127 (3.22) |
| MS24264R22T12PN | .106 (2.69) | .149 (3.78) |
| MS24264R22T19PN | .108 (2.74) | .142 (3.61) |
| MS24264R22T32PN | .109 (2.77) | .152 (3.86) |
| MS24264R22T55PN | .104 (2.64) | .151 (2.16) |
| MS24264R24T43PN | .126 (3.20) | .181 (4.60) |
| MS24264R24T57PN | .125 (3.18) | .178 (4.52) |
| MS24264R24T61PN | .123 (3.12) | .174 (4.42) |

FIGURE 5. Weights for threaded connectors, classes F, G and R.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

| Weight chart | | |
|---|---|---|
| Maximum connector weight in lbs | | |
| Socket insert | | |
| MS PIN | Less contacts | With contacts |
| MS24264R8T2SN | .024 (0.61) | .026 (0.66) |
| MS24264R10T2SN | .040 (1.02) | .042 (1.07) |
| MS24264R10T5SN | .039 (0.99) | .043 (1.09) |
| MS24264R10T20SN | .040 (1.02) | .044 (1.12) |
| MS24264R12T3SN | .055 (1.40) | .060 (1.52) |
| MS24264R12T12SN | .052 (1.32) | .062 (1.57) |
| MS24264R14T3SN | .063 (1.60) | .072 (1.83) |
| MS24264R14T4SN | .059 (1.50) | .074 (1.88) |
| MS24264R14T7SN | .059 (1.50) | .072 (1.83) |
| MS24264R14T12SN | .059 (1.50) | .072 (1.83) |
| MS24264R14T15SN | .059 (1.50) | .072 (1.83) |
| MS24264R16T10SN | .071 (1.80) | .089 (2.26) |
| MS24264R16T24SN | .070 (1.78) | .090 (2.29) |
| MS24264R18T8SN | .084 (2.13) | .112 (2.84) |
| MS24264R18T11SN | .081 (2.06) | .104 (2.64) |
| MS24264R18T14SN | .084 (2.13) | .109 (2.77) |
| MS24264R18T31SN | .083 (2.11) | .108 (2.74) |
| MS24264R20T16SN | .100 (2.54) | .128 (3.25) |
| MS24264R20T25SN | .097 (2.46) | .134 (3.40) |
| MS24264R20T28SN | .097 (2.46) | .131 (3.33) |
| MS24264R20T39SN | .097 (2.46) | .131 (3.33) |
| MS24264R20T41SN | .097 (2.46) | .131 (3.33) |
| MS24264R22T12SN | .111 (2.82) | .153 (3.89) |
| MS24264R22T19SN | .115 (2.92) | .148 (3.76) |
| MS24264R22T32SN | .116 (2.95) | .159 (4.04) |
| MS24264R22T55SN | .110 (2.79) | .155 (3.94) |
| MS24264R24T43SN | .134 (3.40) | .188 (4.78) |
| MS24264R24T57SN | .130 (3.30) | .182 (4.62) |
| MS24264R24T61SN | .128 (3.25) | .178 (4.52) |

FIGURE 5. Weights for threaded connectors, classes F, G and R – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Insert arrangements and alternate insert keying positions shall be in accordance with MIL-STD-1554.

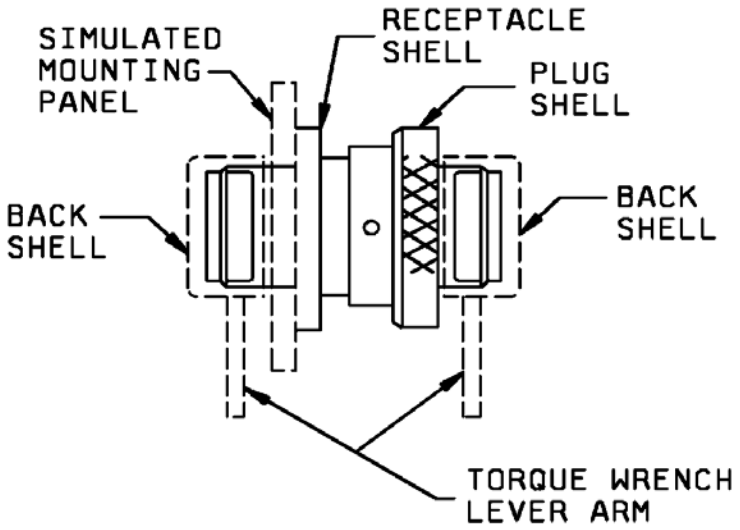
Dimensions and configurations: See figures 1 through 5.

Connector mating: This connector mates with MS24266 and MS27615.

Contacts: In accordance with SAE-AS39029.

For accessories used with this connector, see MIL-DTL-26500.

Accessory thread strength: Connectors with rear accessory threads with optional modified major diameters shall be capable of withstanding the accessory thread strength test. The following details apply: The accessory thread torque for shell size 8 shall be 75 ± 5 inch-pounds. The accessory thread torque for shell size 10 shall be 100 ± 5 inch-pounds. The accessory thread torque for shell size 12 shall be 140 ± 5 inch-pounds. The accessory thread torque for shell sizes 14 through 18 shall be 150 ± 5 inch-pounds. The accessory thread torque for shell sizes 20 through 24 shall be 175 ± 10 inch-pounds. Mated connector pairs shall be mounted as in normal service to a rigid panel. The torque wrench shall be attached as shown in the figure below. After mating the plug and receptacle connectors, a torque shall be applied to the accessory end of the plug at a rate of approximately 10 pound-inches per second until the required torque is achieved. The applied load shall be held for 1 minute, then the load shall be released. The test shall then be repeated on the accessory end of the receptacle.




TEST SET-UP - REAR ACCESSORY THREAD STRENGTH



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

Following accessory thread strength testing, connectors shall be unmated and inspected for damage or breakage through a device having approximately 3X magnification.

Initial qualification and group C requalification (connectors with accessory threads with modified major diameters): Qualification shall be in accordance with MIL-DTL-26500, except the following additional test group shall be required on two mated pairs in each shell size. For initial qualification, accessory thread strength testing may be included in group 1, following the initial visual and mechanical examination:

Visual and mechanical examination.
Accessory thread strength.
Visual and mechanical examination.

Part or Identifying Number (PIN) example:




Application note: The specified torque values are intended for qualification testing only. Using the specified torque values in applications may damage the connectors or equipment.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-DTL-26500, this document references the following:

SAE-AS39029
MIL-DTL-83723
MIL-I-81969/17
MIL-I-81969/19
MIL-STD-1554
MS24266
MS27615

## CONCLUDING MATERIAL

Custodians:
Army - CR
Air Force – 85
DLA – CC

Review activities:
Army - AV
Air Force – 99

Preparing activity:

DLA – CC

(Project 5935–2016-183)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:51Z
Check the source to verify that this is the current version before use.